3.8.2 Magnetic field due to long straight conductor carrying current

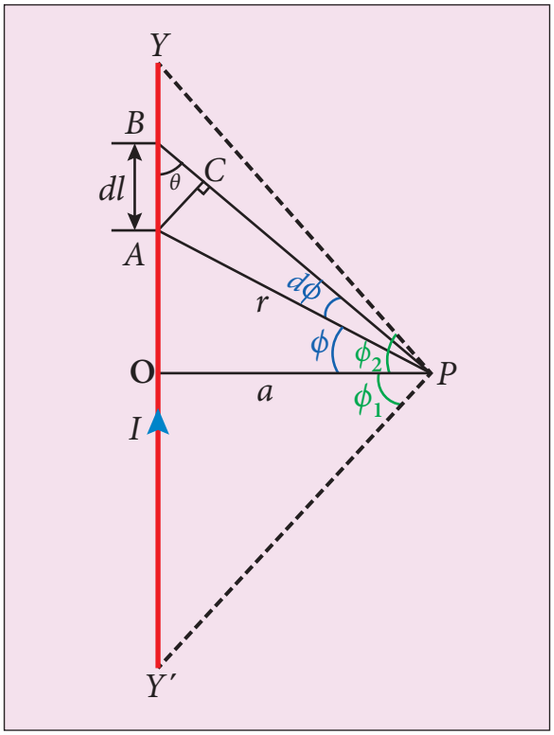


Figure 3.32 Magnetic field due to a long straight current carrying conductor

Let YY′ be an infinitely long straight conductor and I be the steady current through the conductor as shown in Figure 3.32. In order to calculate magnetic field at a point P which is at a distance a from the wire, let us consider a small line element dl (segment AB).

The magnetic field at a point P due to current element Idl can be calculated from Biot-Savart's law, which is

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}\hat{n}$$

where $\hat{n}$ is the unit vector which points into the page at P, θ is the angle between current element Idl and line joining dl and the point P. Let r be the distance between line element at A to the point P.

To apply trigonometry, draw a perpendicular line from A to BP as shown in Figure 3.32.

In triangle ΔABC, $\sin\theta = \frac{AC}{AB}$

$$\Rightarrow AC = AB\sin\theta$$

But $AB = dl \Rightarrow AC = dl\sin\theta$

Let $d\phi$ be the angle subtended between AP and BP

i.e., $\angle APB = \angle APC = d\phi$

In a triangle $\Delta APC, \sin(d\phi) = \frac{AC}{AP}$

Since $d\phi$ is very small, $\sin(d\phi) \simeq d\phi$

But $AP = r \Rightarrow AC = rd\phi$

$$\therefore AC = dl\sin\theta = rd\phi$$

$$\therefore d\vec{B} = \frac{\mu_0}{4\pi}\frac{I}{r^2}(rd\phi)\hat{n} = \frac{\mu_0}{4\pi}\frac{Id\phi}{r}\hat{n}$$

Let ϕ be the angle between AP and OP

In a ΔOPA, $\cos\phi = \frac{OP}{AP} = \frac{a}{r}$

$$\Rightarrow r = \frac{a}{\cos\phi}$$

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I}{a/\cos\phi}d\phi\hat{n}$$

$$\Rightarrow d\vec{B} = \frac{\mu_0 I}{4\pi a}\cos\phi d\phi\hat{n}$$

The total magnetic field at P due to the conductor YY′ is

$$\vec{B} = \int_{-\phi_1}^{\phi_2} d\vec{B} = \int_{-\phi 1}^{\phi_2}\frac{\mu_0 I}{4\pi a}\cos\phi d\phi\hat{n}$$

$$= \frac{\mu_0 I}{4\pi a}\left[\sin\phi\right]_{-\phi_1}^{\phi_2}\hat{n}$$

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